CSA1737 ARTIFICIAL INTELLIGENCE

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① A smart home automation system designs to manage lighting :

* INTRODUCTION :-

An intelligent agent is a system the previous it's environment through the sensors & takes action to achieve desired intelligent agent the control lighting temperature & security based on user behaviour input condition.

TYPES OF INTELLIGENT AGENTS :-

* 1) SIMPLE REFLEX AGENT

1) A simple reflex agent uses pre-defined condition-action rules based on the current input

ADVANTAGES :-

- SIMPLE & FAST
- EASY TO IMPLEMENT

) TRACKS whether a window open before on the air conditioner

- COMBINES door sensors and motion detectors to determine someone maybe home

* ADVANTAGES :-

- Handles incomplete sensor information
- makes more informed decisions

(3) GOAL BASED AGENT :-

A goal based agent selects action by considering a desired goal & determining action can achieve it

ADVANTAGES :-

) can plan action to achieve specific tasks

) more flexible than simple reflex agents.

(2) A robot placed unknown maze & must find path to exit without prior!

* INTRODUCTION :-

A robot placed in an unknown maze must find a path from it's starting position to the exit without knowing maze. detect the walls open and measurements

Two important uninformed research algorithm.

i) uniform cost research (UCS)

ii) iterative reopening search (IDS)

(1) * uniform cost research (UCS)

The uniform cost research explores starting position load and always creates path with cost. it the lowest case first.

* ADVANTAGE :-

* Finds the least-cost path when step costs are non-negative.

* Comparison of Intelligent Agent

Name	Simple reflex	Model-based reflex
Decision making	Current input	Motion-based learning
Planning	No	
Considering	No internal state	
Memory	No	
Learns preferences		
Smart home applications		

* Conclusion :-

A simple reflex agents are useful for rule-based actions while model-based agents maintain information about the environment. Goal-based agents work toward specific objectives and utility-based balance competing comfort, energy and security.

A hybrid intelligent agent combining reflex, model-based, goal-based & utility based decision making with learning mechanism best solutions automation

* COMPARISON BETWEEN UCS AND IDS

• FEATURE	UNIFORM COST RESEARCH
• RESEARCH METHOD	IMPROVEMENT THE LOWEST COST PATH
• USES PATH COST	YES
• TIME COMPLEXITY	EXPONENTIAL IN COMMON-COST CASE FORMULATION.
• SPACE COMPLEXITY	MIN
• BEST SUITED FOR	DIFFERENT MOVEMENT COST.

CONCLUSION :-

THE LEAST COST PATH WHEN MOVEMENT COSTS DIFFER BUT LEAST COST PATH WHEN MOVEMENT COSTS DIFFER ITERATIVE DEEP RESEARCH. IS NOT NECESSARY. AND REMOVE ALL MOVEMENT HAVE EQUAL COST.

2) Iterative Deepening search :-

Iterative Deepening research combines the depth - first approach with breadth later depth limits it gradually increases the max depth is found.

* ADVANTAGES :-

- Reactive lev MEMORY than used
- 1 complete when the branching factor finite
-) useful when the maze depth is unknown
-) find a shortest solution when all actions have equal cost.

* Time & space complexity :-

-) Time : O(b^d)
-) space : O(bd)

Here :-

-) b = branching factor
-) d = depth solution